Treffen der Kooperationsinitiative

TU Braunschweig

Medizinische Hochschule Hannover

Städtisches Klinikum Braunschweig

und Kooperationspartner

22.04.2026

Steckbriefe

Technische Universität Braunschweig

skbs

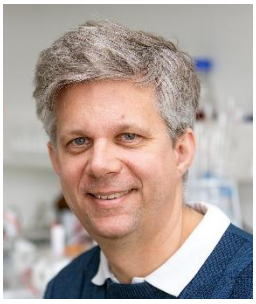
STÄDTISCHES KLINIKUM BRAUNSCHWEIG

Medizinische Hochschule Hannover

Steckbriefe

Medizinische Hochschule Hannover

Systems Biology of Herpesvirus Infections – Hannover Medical School



Name: Prof. Dr. med. Lars Dölken

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Research Topics

model organism: **Man/Mouse** **model tissue:** **Brain, skin, gut, cell lines**

Infection models: **Human and murine herpesviruses (HSV-1, HCMV, MCMV)**

- molecular mechanisms of host–virus interactions (integrative multiomics)
- viral manipulation of the host transcriptional machinery (Pol II)
- time-resolved single-cell transcriptomics and transcriptional dynamics in viral infections
- cellular stress responses in viral infection and sensing virus-/stress-induced cellular long aberrant nuclear RNAs arising from disrupted Pol II transcription termination
- molecular determinants of viral latency–lytic switch
- systems-level decoding of herpesvirus genomes

Research Techniques

- time-resolved single-cell RNA-seq (scSLAM-seq) and metabolic RNA labeling
- integrative transcriptomics, translatomics (Ribo-seq), and proteomics
- viral reverse genetics and mutagenesis approaches
- primary cell infection models
- lentiviral gene delivery
- genome-wide and targeted CRISPR screens
- bridging wet-lab and bioinformatics for integrative multiomics analysis

Collaboration wishes

- *in vivo* confocal microscopy, *in vivo* cell biology, optogenetics
- primary cell culture, iPSC and organoid models, on-chip approaches
- novel gene delivery approaches
- new technologies, e.g., for cell manipulation, drug delivery, compound studies, and imaging

Biologische Grundlagen für Biohybridimplantate - Medizinische Hochschule

Prof. Dr. rer. nat. Andrea Hoffmann

Klinik für Orthopädie
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Keine Teilnahme am 22.04.2026

Research Topics

model organism: **human**, clinical

model tissue: **bone marrow, blood, synovial fluid / musculoskeletal system**

- Body´s own stem cells (mesenchymal stem cells) and immune cells (monocyte-derived macrophages) for optimized performance of implants/biomaterials: isolation and cultivation with reduced oxygen, mimicry of stem cell niche
- Fibrosis: a severe adverse tissue reaction with excessive extracellular matrix formation and dysfunction of immune cells
- Hard tissue sarcoma: altered functions of stem cells
- Aging: altered functions of stem cells

Research Techniques

- Genetic (lentiviral) modification of cells
- OMICs, e.g. correlation of features identified by RNA-sequencing with live cell isolation by flow cytometry
- Cell culture with reduced oxygen
- Signal transduction in differentiation and inflammation

Collaboration wishes

- Metabolome alterations in sarcoma / under hypoxia (physioxia)
- Microfluidic devices for hypoxic isolation of stem cells
- 3D dynamic cell culture
- 3D substrates for cell culture (hydrogels, scaffolds,….)
- Low molecular weight substances with biological activity
- Bioinformatics / computational biology for -OMICs data interpretation

Applied Macrophage and Stem Cell Research - Hannover Medical School

Prof. Dr. Nico Lachmann

Clinic for Pediatric Pneumology, Allergology and Neonatology
Hannover Medical School

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Research Topics

"model" organism: human **model tissue:** hematopoietic cells

- Synthetic biology approaches developing cell-based immunotherapies
  Design of artificial receptor, macrophage therapies for lung regeneration, infection

- Generation of human hematopoietic organoids producing human immune cells
  Tissue resident macrophages, innate immune cells, functional characterization,
  Scaleable differentiation systems, differentiation techniques and technologies

- Pathways and targets modulating inflammation in human innate immune cells
  Drug screening, identification of ncRNAs, modulating macrophage function,
  NLRP3 targets, reporter systems, bioassays for pro/anti-inflammation

Research Techniques

- Generation, handling and differentiation of (patient) human induced pluripotent stem cells and hematopoietic stem cells towards innate immune cells

- Polarization of human macrophages and expertise in the generation of tissue macrophages also from different genetic backgrounds (patients)

- Scalable production of human hematopoietic organoids, bioreactors and process development

- Genetic enhancement (e.g. lentiviral vectors) to manipulate human hematopoietic cells, GFP reporter assays and live-cell imaging techniques

Collaboration wishes

- Design of "tissue on a chip" prototypes integrating human macrophages to study role and function of macrophage in e.g. brain/liver/lung for drug development

- Developing drug screening pipelines applying patient immune cells, modulating inflammation (pro/anti) > automization approaches using e.g. chip designs

- Techniques and approaches which are able to manipulate the fate of macrophages in vivo (e.g. controlled release of compounds/drugs in vivo, or controlled apoptosis of cells in vivo) together with HZI

Department of Plastic, Aesthetic, Hand and Reconstructive Surgery-Hannover Medical School

Prof. Dr. med. Peter M. Vogt

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Standort: Hannover Medical School

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Research Topics

model organism: **humanized mouse models** **model tissue:** **human skin grafts**

- Immunological mechanisms of skin graft rejection
- MHC class I mediated antigen presentation
- T cell mediated immune responses in transplanted skin
- Development of novel immunomodulatory therapies
- viral immune evasion proteins (e.g., US11)
- local modulation of immune responses in transplanted tissue
- Nanoparticle-based drug delivery systems
- Niosome based protein transport systems
- topical formulations (cream / hydrogel)
- controlled release and dermal penetration of therapeutic molecules
- Translational research in reconstructive medicine
- preclinical models for skin transplantation
- preclinical efficacy and safety studies
- preparation of early clinical trials

Research Techniques

- development and characterization of nanoparticle-based delivery systems (niosomes)
- protein analytics (Western blot, ELISA, chromatographic methods)
- *ex vivo* human skin penetration studies (Franz diffusion cells)
- in vivo studies using humanized mouse models
- histology and immunohistochemistry of skin tissue
- molecular analyses and flow cytometry (RT-qPCR, immune markers)
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Collaboration wishes

- clinical collaborations in burn injury and skin transplantation
- interdisciplinary collaborations in (nanoparticle-based) drug delivery and biomaterials
- interdisciplinary collaborations in oncology
- partnerships for translational development and regulatory preparation of
- innovative biopharmaceutical therapies and biomaterials (spider silk)

Steckbriefe

Städtisches Klinikum Braunschweig

Städtisches Klinikum Braunschweig gGmbH - skbs

Thomas Bitter

Klinik für Pneumologie und Beatmungsmedizin
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Research Topics

model organism: **model tissue:**

1.) Schlafmedizin:

Erforschung von klinischen und pathophysiologischen Zusammenhängen zwischen schlafmedizinischen Pathologien (Obstruktive Schlafapnoe, Cheyne-Stokes Atmung, Insomnien etc.) und internistischen (insbes. Kardiopulmonale) Erkrankungen

2.) Interventionelle Pneumologie
Erforschung von neuen Techniken zur verbesserten Diagnostik peripherer Lungenrundherde, interventionellen Therapie der fortgeschrittenen COPD sowie individualisierte Stentversorgung
3.) Telemedizin
Erforschung des Einsatzes von Telemedizin zur Früherkennung und Sekundärprophylaxe und klinischen und prognostischen Verlaufsbeurteilung pneumologischer Erkrankungen
4.) Infektiologie
Erforschung verbesserter Diagnostikmöglichkeiten der Tuberkulose (in Kooperation mit Prof. Dr. Mahavir Singh, Lionex GmbH in Braunschweig)

Research Techniques Klinische Forschung

Collaboration wishes

1.) Interventionelle Pneumologie:
   a. wir würden uns über eine Partnerschaft zur Entwicklung einer im Vergleich zu bestehenden Systemen vereinfachten und präziseren KI-basierten Navigation zu peripherin Rundherden freuen.
   b. Gemeinsame Entwicklung Individualisierter 3D-Stents bei trachealen und bronchialen Stenosen
2.) Telemedizin
   a. Entwicklung eines intelligenten Closed-Loop O2-Systems zur bedarfsadaptierten O2-Gabe bei respiratorisch insuffizienten Patienten
   b. Entwicklung kontaktloser Devices zur verbesserten/frühzeitigen Vorhersage akuter Exazerbationen bei Patienten mit z.B. COPD oder interstitiellen Lungenerkrankungen
3.) Schlafmedizin
   a. Evaluation der klinischen und prognostischen Bedeutung verschiedener schlafmedizinischer Parameter (Schlafdauer, Insomnie, Hypersomnie etc.) bei Patienten mit pneumologischen Krankheitsbildern

Städtisches Klinikum Braunschweig gGmbH - skbs

Ulrike Diedrich

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Aktuelle Tätigkeit und Qualifikation

Abteilungsleitung Klinische Pharmazie und Stationsapotheker

Fachapothekerin für Klinische Pharmazie, Pharmazie in der Geriatrie und Medikationsmanagement im Krankenhaus

Projektleitungsteam Einführung Verordnungssoftware

Bisherige Kooperation:

TU BS: Lehrauftrag für eine Vorlesung im 8. Semester für die Pharmaziestudierenden: Curriculum Klinische Pharmazie

Interesse an:

- Austausch zu möglichen weiteren Lehrtätigkeiten an der TU BS
- Möglichkeiten der berufsbegleitenden Promotion?
- Austausch mit MHH zur Einführung der Verordnungssoftware / Implementierung eines neuen KIS

Städtisches Klinikum Braunschweig gGmbH - skbs

Dr. rer. nat. Cathrin Klopp

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Tätigkeiten

- Stationsapothekerin
- Fachapothekerin für Pharmazeutische Analytik und Technologie
- In Weiterbildung zur Fachapothekerin in dem Gebiet Klinische Pharmazie sowie dem Bereich Medikationsmanagement im Krankenhaus

Promotion:

- CAU Kiel: Arbeitsgruppe Strukturbiologie (Prof. Dr. Axel J. Scheidig) und Pharmazeutische und Medizinische Chemie (Prof. Dr. Bernd Clement)
- Thema: Neue analytische Verfahren zum Nachweis von mARC-Substraten

Kooperationswünsche

- Interprofessioneller Austausch zu den Themen Biotransformation/Arzneistoffmetabolismus, Therapeutisches Drug Monitoring und Genotypisierung

Städtisches Klinikum Braunschweig gGmbH - skbs

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Tätigkeiten

Stationsapothekerin

Fachapothekerin für Klinische Pharmazie und

Pharmazie in der Geriatrie

Fachapothekerin für Offizinpharmazie und Ernährungsberatung

Ehrenamtliche Tätigkeit

Vizepräsidentin der Apothekerkammer Niedersachsen

Aktuelle Kooperationen

PJ – Akademie Kooperationsvereinbarung skbs/ TUBS/ Apothekerkammer Niedersachsen

Kooperationswünsche

Ausbau der interprofessionellen Zusammenarbeit

Institut für Klinische Transfusionsmedizin (IKTM) – Städtisches Klinikum BS

Dr. Henk Garritsen/ Prof. Dr. Thomas Müller

IKTM
SKBs S3 (Celler Str. 38)

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Research Topics

model organism: **human** (patients, healthy donors)

model tissue: blood stem cells ($CD34^+$), platelets, PMN, cornea, large arteries/aorta bone

- survival of (modified) donors cells in recipient blood
- cell-cell and cell–matrix Interactions
- cell growth and cell differentiation
- HLA-Typing

Research Techniques

- next generation sequencing
- quantitative PCR
- cell sorting and flow-cytometry
- cell culture
- HLA-Typing

Collaboration wishes

- survival of transfused/transplanted modified cells in recipients
- functional models of blood – vessel wall interaction
- tools to assess prothrombotic states

Städtisches Klinikum Braunschweig (SKBS)

Prof. Dr. med. Philipp Wiggermann | Dr. rer. nat. Carolina Rio Bartulos
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Research Topics

Image-guided oncology and minimally invasive therapy: development and clinical evaluation of thermal and non-thermal ablation, precision assessment, robotics, treatment response evaluation and workflow optimization in oncologic care.
Quantitative imaging, AI and imaging biomarkers: MRI-, CT-, US- and PET-based biomarkers for diagnosis, risk stratification and therapy monitoring; integration of radiomics,
machine learning and workflow automation into clinical imaging.
Neuroimaging, molecular imaging and radiation safety: stroke imaging and neurointervention, PET/CT and tracer-based diagnostics, as well as dosimetry, dose reduction, quality assurance and radiation incident management.

Research Techniques

- Multimodal imaging acquisition and quantitative analysis (MRI, CT, PET/CT, ultrasound, angiography).
- Image-guided interventional and robotic-assisted procedures with prospective clinical evaluation.
- AI, machine learning and radiomics for image analysis and workflow optimization.
- Radiation dosimetry, risk assessment and quality assurance.

Collaboration wishes

- Joint development and clinical validation of AI-based imaging biomarkers and workflow tools using clinically annotated SKBS datasets, together with the expertise of future research partners in data science, modelling and engineering.
- Collaborative projects in image-guided and robotic-assisted intervention for oncologic and neurovascular applications, including navigation, precision assessment, workflow integration and safety evaluation.
- Translational cooperation in molecular imaging, tracer evaluation and dosimetry, linking research partners' strengths in chemistry, physics, life sciences and medical technology with PET/CT, interventional radiology and nuclear medicine at SKBS.
- Establishment of a joint translational imaging and clinician-scientist platform as a nucleus for future medical faculty structures in Braunschweig.

Städtisches Klinikum Braunschweig gGmbH – skbs

Name: Silvia Varotto
Medizinische Klinik IV- Geriatrie (80 Betten mit alterstraumatologisch, neurologisch und internistisch akuterkrankten Patientinnen und Patienten)
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Research Topics: KEINE

model organism: **model tissue:**

Research Techniques: KEINE

Collaboration wishes

- collaboration proposal 1

Klinische Aspekte der Medizininformatik: App geriatrisches Assessment (Erfassung der Selbsthilfefähigkeit, Mobilität, kognitive Störung, Umgang mit mobilen Geräten im Alter, etc.)

- collaboration proposal 2

Verbindung ingenieurwissenschaftlicher Ansätze/ Gerontotechnologie und Klinik: Sturzanerkennungssensoren, Robotik, Vereinfachte Tablets oder Videokommunikationssysteme, Virtuelle Realität (VR) zur kognitiven Stimulation, tragbare Sensoren etc.

- collaboration proposal 3

Verbindung Pharmazie und Geriatrie: Risikoreduktion bei Polypharmazie, Arzneimitteltherapiesicherheit (bereits etablierte Kooperation mit Klinikapotheke)

Steckbriefe

TU Braunschweig

und Partner

Medical Informatics - Technische Universität Braunschweig

Diana Baumgärtel

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Research Topics

Continuous Health Monitoring:

- Using ambient smart home sensors to capture environmental, behavioral, physiological, and psychological parameters, aiming to derive behavioral and sleep patterns and assess their relationship with health and well-being
- Signal processing and quality assessment for biosignals (e.g., electrocardiography)

Research Techniques

- Data analytics on time-series, videos, and multimodal sensor data
- Signal processing and preprocessing
- Deterministic algorithms and explainable AI approaches

Collaboration wishes

- Case studies
- Clinical trials
- Teaching

Name Prof. Dr. med. Sönke Behrends (Facharzt Klinische Pharmakologie)

Institut Pharmakologie
Standort Mendelssohnstrasse 1, 38106 Braunschweig

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Research Topics

model organism: Mensch, Maus **model tissue:** Herz, Hirn, Lunge, Ovar

Struktur und Funktion der NO-sensitiven Guanylatzyklase und Charakterisierung neuer Stimulatoren und Aktivatoren in Kooperation mit Bayer

Vericiguat bei Herzinsuffizienz (einschließlich Doxorubicin-induzierter)

Humane GUCY1A2-Gain-of-Function-Mutation mit McCune-Albright-ähnlichem endokrinem Phänotyp (cGMP–cAMP Crosstalk)

Evolution des humanen GUCY1-Tandems: Bystander-Gene und HCNEs

Research Techniques

- Klassischer Enzymassay, Aufreinigung aus Sf9-Zellen, Thermostabilität
- Konfokale Mikroskopie, FRET
- Mausmodell der GUCY1A2-Gain-of-Function-Mutation (und analoges GUCY1A1-Gain-of-Function-Mausmodell)
- Rattenmodell der Doxorubicin-induzierten Herzinsuffizienz in Kooperation

Collaboration wishes

- Doxorubicin-induzierte Kardiomyopathie und Vericiguat
- Ovarielle Hyperfunktion durch cGMP/cAMP-Erhöhung in Granulosazellen
- Gemeinsame Betreuung medizinischer Doktorarbeiten (Bystander / HCNEs)

Virology and Innate Immunity - Technische Universität Braunschweig

Prof. Dr. Melanie Brinkmann

Institute of Genetics
Technische Universität Braunschweig

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Research Topics

model organisms: organoids, mice

focus pathogens: Herpesviruses, Coronaviruses

- innate immune response to viral infections; zoonotic risks
- viral modulation of the innate immune response
- identification and molecular characterisation of viral proteins essential for establishment of lifelong chronic infection
- identification and molecular characterisation of cellular processes supporting chronic/persisting infection
- role of bone morphogenetic proteins as novel players of the innate immune response for control of viral infections

Research Techniques

- tracking of viral particles and viral protein expression in tissue culture (live cell imaging)
- gene and protein expression dynamics during viral infection (RNAseq, proteomics)
- genetic manipulation of viral genomes

Collaboration wishes

- clinical cases in the area of neurodegeneration: longterm effects of herpesviral infections
- clinical cases of herpesviral caused encephalitis
- tracking of viral infections
- primary cells from patients; lung explants

Institut für Pharmazeutische Technologie und Biopharmazie - Technische Universität Braunschweig

Prof. Dr. Bunjes, Heike

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pvz Center of Pharmaceutical Engineering

Research Topics

- Nanoparticulate drug delivery systems for poorly water-soluble drug substances
  Lipid-based nanoparticles (solid lipid nanoparticles, nanoemulsions, liposomes, dispersions of liquid crystalline phases); drug nanoparticles; structure-function relationships in such systems (regarding, e.g., stability, drug loading and release); novel preparation processes and effects of pharmaceutical process steps (e.g. dispersion processes, sterilization) on nanoparticle properties)
- Formulations for therapeutic biomacromolecules
  Development and characterization of injectable/implantable formulations for controlled protein release; strategies for the administration of therapeutic biomacromolecules (proteins, RNA) using lipid nanoparticles
- Self-dispersible and patient-individualized drug dosage forms
  self-dispersible lipid and lipid-loaded hydrogel-based pellets

Research Techniques

- High- und low-energy-emulsification; microfluidic precipitation; freeze and spray drying; hot-melt extrusion, prilling
- Particle size analysis (dynamic light scattering, laser diffraction, nanoparticle tracking analysis, dynamic image analysis)
- Differential scanning calorimetry (DSC) and isothermal microcalorimetry
- Polarized light microscopy
- Small- and wide-angle X-ray diffraction
- Interfacial tensiometry, contact angle analysis
- Electron microscopy

Collaboration wishes

- Demanding applications requiring development of efficient delivery systems for poorly soluble or sensitive biomacromolecular (e.g., RNA, peptides, proteins) active substances
- Approaches requiring patient-individualized drug dosage forms

Medical Informatics - Technische Universität Braunschweig

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Keine Teilnahme am 22.04.2026

Research Topics

Continuous Health Monitoring:

- Smart environment (Smart home, smart car, …)-integrated sensing of vital signs and other environmental, behavioral, physiological, and psychological parameters
- Sensor design and signal validation

Early Rescue Chain:

- International Standard Accident Number (ISAN)
- Digital Rescue Sheet

One Digital Health:

- Combining human, environment, and animal data
- Data transfer and cross-species learning

Research Techniques

- Data analytics and explainable artificial intelligence (xAI)
- Terminology and event coding with ICD11
- System prototyping/demonstrator building

Collaboration wishes

- Case studies
- Clinical trials

Microbial Proteomics - TU/HZI

Prof. Dr. Susanne Engelmann

Institute for Microbiology
Technische Universität Braunschweig
Microbial Proteomics
Helmholtz Center for Infection Research

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Research Topics

model organism: *Staphylococcus aureus, Pseudomonas aeruginosa*

- studying the dynamics of the protein inventory, protein-protein interactions and protein modification events to understand the bacterial physiology under infection relevant conditions and in bacterial communities
- identification and characterization of essential bacterial proteins (with the special focus on small proteins and proteoforms) that serve as novel drug targets
- role of bacterial ribosome heterogeneity in antibiotic resistance and during host pathogen interaction

Research Techniques

- physiological proteomics (including bacterial panproteomics)
- proteogenomics (including N-terminomics) for identification and characterization of small proteins and proteoforms
- ribosome präparation
- metaproteomics
- interactomics
- expression and purification of proteins, mutant construction

Collaboration wishes

- medical relevant microbiomes with special focus on nasal microbiomes

Institut für Partikeltechnik - Technische Universität Braunschweig

Dr. Jan Henrik Finke

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Research Topics

Modellorganismus: Saccharomyces cerevisiae
Modellgewebe: Jegliches Gewebe, das Ziel der Arzneimittelanwendung, einer Transportbarriere oder der Arzneimittelwirkung ist

- **Spezifische Darreichungsformen:**
  Entwicklung von Arzneiformen, die auf spezifische Applikationsorte, Wirkstofffreisetzungsverhalten und Herausforderungen des Wirkstoffs (z. B. geringe Löslichkeit, Empfindlichkeit) zugeschnitten sind.
- **Effiziente Produktionsprozesse:**
  Forschung an Herstellungsprozessen
  Entwicklung von Modellen zur Vorhersage von Struktur, Eigenschaften und Performance der Darreichungsformen
- **Individuelle und patientenzentrierte Produktion:**
  Prozesse zur sicheren Herstellung von Arzneiformen mit individueller Dosierung
  Kombination von Wirkstoffen und kontrollierte Freisetzungsprofile

Research Techniques

- Untersuchung von Produktions- und Herstellungsprozessen von Darreichungsformen
- Formulierung sensibler biologischer Moleküle oder ganzer lebender Zellen (z. B. RNA, Proteine, Probiotika)
- In-vitro-Charakterisierung pharmakologischer, physikochemischer und mechanischer Eigenschaften auf verschiedenen Skalen
- Raman-Mikroskopie zur chemischen Kartierung
- Kryo-SEM zur in situ-Analyse von vitrifizierten Proben (z. B. Zellen, Dispersionen)
- Nanopartikel-Forschung:
- Herstellung und Charakterisierung von Wirkstoff-Nanopartikeln, Lipid-Nanopartikeln, Emulsionen
- Trocknung von Nanopartikeln zu festen Produkten

Collaboration wishes

- Untersuchung von Produktions- und Herstellungsprozessen von Darreichungsformen
- Formulierung sensibler biologischer Moleküle oder ganzer lebender Zellen (z. B. RNA, Proteine, Probiotika)
- In-vitro-Charakterisierung pharmakologischer, physikochemischer und mechanischer Eigenschaften auf verschiedenen Skalen
- Raman-Mikroskopie zur chemischen Kartierung
- Kryo-SEM zur in situ-Analyse von vitrifizierten Proben (z. B. Zellen, Dispersionen)
- Herstellung und Charakterisierung von Wirkstoff-Nanopartikeln, Lipid-Nanopartikeln, Emulsionen
- Trocknung von Nanopartikeln zu festen Produkten

Particle Technology / Nanomaterials - Technische Universität Braunschweig

Prof. Dr. Georg Garnweitner

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Research Topics

- **Multifunctional Niosomes**: Developing specific vesicles acting as a modular platform for simultaneous drug delivery, magnetic targeting, and diagnostic imaging.
- **Tailored Aerogels for Drug Delivery**: Synthesis of highly porous silica structures designed to stabilize amorphous drugs and improve the dissolution of poorly soluble APIs.
- **Magnetic Nanoparticles for Downstream Processing and Cell Sorting / Diagnostics**: Design of functionalized magnetic particles/beads for the efficient high-capacity separation of metabolites (e.g. proteins) and cells from complex biotechnological media as well as for diagnostic applications.
- **Lipid-Nanoparticle Formulations**: Engineering of advanced lipid-based carrier systems specifically for the stabilization and targeted delivery of mRNA and other sensitive biopharmaceuticals.

Research Techniques

- **Innovative Synthesis Routes (e.g. Nonaqueous Sol-Gel Synthesis)**: Various "bottom-up" approaches in organic/aqueous solvents that allows for precise control over the crystallinity, size and shape of nanoparticles.
- **Surface Modification and Functionalization**: Targeted multi-step modification strategies of the particle surface to ensure stability in biological fluids and enable the attachment of targeting ligands.
- **Preparative Gel Electrophoresis**: Utilization of advanced separation techniques to purify and fractionate nanoparticles by size or charge.
- **Advanced Characterization (Nanoanalytics, e.g. SAXS & TEM)**: Application of Small-Angle X-ray Scattering and high-resolution electron microscopy to analyze particle formation and morphology in detail; analytical method development.
- **Crystallization and Crystal Engineering**: Targeted design of crystals and morphologies as well as interfaces to optimize the magnetic, optical, or catalytic properties of nanomaterials.

Collaboration wishes

- Novel APIs for formulation with our carrier systems
- In-vitro and in-vivo applications
- Possible applications: MRI, MPI, hyperthermia, fluorescence, drug delivery

Prof. Dr. Andriy Goychuk

Braunschweiger Zentrum für Systembiologie
Helmholtz-Zentrum für Infektionsforschung / TU Braunschweig

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Research Topics

- Chromatin Organization
- Biomolecular Condensates
- Mechanobiology
- Stochastic Dynamics of Infection Processes in Tissue

Research Techniques:

- Theoretical Physics: Statistical Mechanics, Field Theory, Stochastic Processes, Transport Processes, Reactions
- Engineering: Finite Element and Finite Volume simulations, Molecular Dynamics simulations
- Mathematical Biology: Pen & Paper calculations

Collaboration wishes:

- Systems level biological systems (i.e., interactions between glucose metabolism and immune response in infection and sepsis)
- Within-host infection dynamics
- Anything that makes us think “This is weird. I don’t understand it.”

Abteilung Medizinischen Biotechnologie - TU Braunschweig

Prof. Dr. Michael Hust

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TU Braunschweig

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Research Topics

Research Techniques

- Antibody engineering and antibody generation (phage display)
- Biomarker identification (ORFeome phage display)
- CAR-T cells (several cooperations MHH, Charité...)
- Neutralizing anti-viral/toxin antibodies

Collaboration wishes

- Clinical partner needing engineered human antibodies for diagnostic and new therapeutic concepts

Cellular & Translational Proteomics - Helmholtz Centre for Infection Research

Prof. Dr. Lothar Jänsch

Department Structure and Function of Proteins
Helmholtz Centre for Infection Research

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Research Topics

organism: human **tissue:** mucosal immunity

- Functional phenotyping of blood and tissue-resident mucosal immune cell types (**MAITs**, **monocytes** and **macrophages**).
- Characterize immune cell activation pathways and donor-specific immune responses to **infection** and **tissue injuries**.
- Develop analytical strategies to monitor extracellular (secreted) immune signaling **proteoforms** in body fluids.
- Identify **surface receptors / markers** of infection-adapted immune cell subsets.
- **Virus-host interfaces** (glycosylation signatures of MERS-CoV, Marburg virus and host entry receptors)

Research Techniques

- Ultra-sensitive targeted and untargeted proteomics of primary materials (BSL1-4).
- Spatial proteomics (tissues, cells, protein complexes) by laser-dissection microscopy and interactomics.
- Flow cytometry / antibodies for isolation/definition and polarization of primary immune cell subsets.
- Surfaceomics / glycoproteomics of immune cell subsets, viruses, antibodies.
- Phosphoproteomics for signal network status and dynamics (2 - 30min).
- Biorthogonal non-canonical amino acid tagging (BONCAT) for cell-specific translatome and secretome studies (*ex vivo, in vivo*).

Collaboration wishes

- Access to tissue lesions and histology to validate results from our *ex vivo* MAIT cell wound healing phenotypes (skin, intra-/extra-hepatic bile duct).
- Development of immune-competent iPSC-derived organoid and barrier models: gut-bile-liver; nose-brain; blood-brain. - Study MAIT cells from patients with (i) type 2 diabetes and (ii) before & after antiviral therapies.

Data Science in Biomedicine – TU Braunschweig

Prof. Dr. Tim Kacprowski
Institute of Data Science in Biomedicine
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Research Topics

model organism: **model tissue:**

-*Systems Medicine*
patient stratification, disease subtyping

- *Network Biology*
molecular interaction networks, regulatory networks, knowledge graphs, network inference

- *Genetic Epidemiology*
genome-wide association studies, polygenic risk-scores, epistasis

- *Multi-Omics & Data Integration*

Research Techniques

- Machine Learning / Artificial Intelligence
- (Bio-) Statistics
- Graph Theory
- Algorithmics

Collaboration wishes

- patients and clinical data for neurological or neurodegenerative diseases
- patients and clinical data for other diseases
- molecular and genetic data
- time-series data

Institut für Thermodynamik - Technische Universität Braunschweig

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pvz
Center of Pharmaceutical Engineering

Research Topics

model organism: **model tissue:**

- nasal microbiome (mucosa) and olfactory epithelium
- key microbial species (principal metabolic products) in the nasal microbiome, e.g., bacteria, phages, virus surrogates (nanoparticles), fungi
- formulated active ingredients
- focus on systems in liquid phase without or with dispersed particles (additional option to address gaseous phase)

Research Techniques

- light scattering techniques (dynamic light scattering, Raman spectroscopy) for study of diffusive mass transport (diffusion coefficients), particle morphology, and fluid concentration
- drop analysis methods (pendant drop, sessile drop) for analysis of interfacial tension and wetting effects
- transient hot-wire method for measurement of thermal conductivity down to small volumes in µL scales
- additional expertise in molecular dynamics simulations for studying molecular and macroscopic properties as well as in development of semi-empirical relationships between molecular structure and macroscopic fluid properties

Collaboration wishes

- collaborations with the institutes embedded within PVZ, BRICS, and HZI on the study of model or real molecular and particulate systems in pharmeutical engineering and microbiology
- interest in method transfer to systems relevant for clinics and ophthalmologists for study of transport and interfacial properties
- partnerships with pharmaceutical companies for providing useful information on the design of particulate products

Cellular and Molecular Neurobiology - Technische Universität Braunschweig

Prof. Dr. Martin Korte

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AG Neurodegeneration and Neuroinflammation

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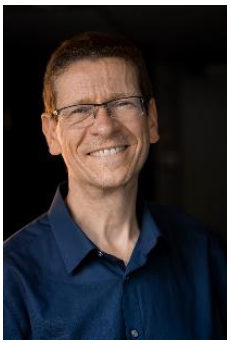

Research Topics

model organism: mouse

model tissue: Hippocampus, Cortex, Amygdala

- molecular and cellular mechanisms of synaptic plasticity, learning, memory and forgetting
- infection models (viral and bacterial) and sepsis and effects on the brain (neuroinflammation)
- investigation of genetic mouse models of neurodegenerative diseases:
- Alzheimer Disease (AD), Parkinson disease
- Developmental diseases: FXS, Autisms-Spectrum disorder
- Psychological disease: depression and chronic stress
- Development of new biomedical instruments

Research Techniques

- *in vivo* confocal microscopy, *in vivo* imaging, STED (high resolution) microscopy
- electrophysiology (extracellular recordings, patch-clamp)
- primary cell culture, neurons on (microfluidic) chip, microglia cultures, tri culture systems, organoids from human IPSC cells.
- Ca^{2+}-imaging
- Behavioral experiments with mice

Collaboration wishes

- clinical cases and therapeutic approaches for infections: from lung, blood and brain (MRT, PET)
- clinical cases of neurodegeneration
- blood samples for the analysis of cytokines, chemokines and neurotrophic factor under disease conditions
- chemobrain: better understanding of neurinflammation and its consequences after chemotherapy or radiotherapy
- functionalized chambers for multimodal characterization (electrophysiological,
- microscopy, respiration) of primary neuronal cultures, comparison to clinical
- samples
- analysis of structure, pharmacokinetics and metabolism of possible anti-inflammatory mitochondrial products or drug compounds

Cellular and Molecular Neurobiology - Technische Universität Braunschweig

Prof. Dr. Reinhard Köster

Institute for Cell and Neurobiology
Technische Universität Braunschweig

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Research Topics

model organism: zebrafish **model tissue:** cerebellum

- molecular and cellular mechanisms of neuronal differentiation (in the cerebellum), neuronal migration, axon pathfinding, autophagy, synaptogenesis,
- investigation of genetic models of human neurodegenerative diseases: Spinocerebellar Ataxias (SCAs 1, 2,13), Alzheimer's Disease (AD), motoneurons, protein aggregation dynamics, neuroinflammation, Dyrk1A-kinase
- mechanisms of neuronal regeneration in the cerebellum, behavioral analysis
- development of genetic and imaging techniques

Research Techniques

- genetic manipulation of zebrafish
- *in vivo* confocal microscopy, *in vivo* cell biology, optogenetics
- primary cell culture, neurons on (microfluidic) chip, zebrafish on chip (Ca^{2+}-imaging)
- generation of pseudotyped Baculovirus for primary cell culture infections
- nanoparticle (carbon quantum dots) and compound studies *in vitro* and *in vivo*

Collaboration wishes

- clinical cases and therapeutic approaches for cerebellar diseases
- functionalized chambers for multimodal characterization (electrophysiological, microscopy, respiration) of primary neuronal cultures, comparison to clinical samples
- analysis of structure, pharmacokinetics and metabolism of quantum dots, usefulness as contrast agents (MRI, PET?)

Physikalische Chemie Biomolekularer Systeme – Technische Universität Braunschweig

Prof. Dr. Jacek Kozuch

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Research Topics

model systems/systems of interest: isolated proteins, viruses, tissues for imaging

- characterization of molecular origins of evolution of antibiotic resistance (currently: class A β-lactamases)

- characterization of virus-membrane interactions and viral proteins (currently: influenza A viruses and proteins)

- protein misfolding at membrane interfaces

- activity testing of inhibitors etc.

Research Techniques

- surface-sensitive infrared spectroscopy in combination with model membrane systems (*surface-enhanced infrared absorption spectroscopy*)

- combinations of atomic force microscopy and infrared spectroscopy – label-free imaging with ~20 nm resolution (*scattering-type scanning nearfield optical microscopy* and *nanoscale-FTIR spectroscopy*)

- molecular dynamics simulations with higher-level force fields (e.g. polarizable AMOEBA force field)

- heterologous protein expression; introduction of noncanonical amino acids via amber suppression

Collaboration wishes

Note: label-free AFM-IR-imaging is still in development phase

- label-free imaging of tissues, e.g. with misfolded proteins

- tracking of viral particles

Fraunhofer IST

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pvz Center of Pharmaceutical Engineering

Research Topics

model organism: any organism (S1 possible at IST), where interaction with surfaces or with cold plasma plays a role

model tissue: any tissue, where enhanced adhesion or repellent surfaces are of interest, moreover plasma-tissue interaction may play a role

Surface modification for medical technology

- **Surface functionalization and coating** for medical devices and implants for improved **adhesion or anti-adhesive** applications focused on thin film technology, e.g. **plasma technology**
- **Cleaning and disinfection/sterilization** by sustainable technologies (plasma activated water, electrochemical systems)
- **Area-selective Microplasmas** or modification of **microfluidics**

Surface modification for pharmaceutical applications

- Modification of **particles** for pharmaceutical engineering by plasma technology
- Primary packaging
- Wear-resistant and anti-adhesive coatings for **production tools**

Research Techniques

- Plasma-enhanced chemical vapor deposition (PECVD), PVD, ALD
- Microplasmas
- Surface analysis and characterization (FTIR, Raman, SIMS, AFM, XPS,…)

Collaboration wishes

- Clinical cases, demands, and therapeutic approaches
- Development of new devices or packaging with improved drug delivery for standard or novel drug systems
- Waste reduction / improved sustainability in clinical surrounding → improved packaging or sterilization techniques, novel materials, design and functionality of PoC or IVD,…
- Integration of plasma-based
- Any challenges where surface interaction is of interest (adhesion, anti-sticking, antimicrobial, release,…)

Applied Microbiology – Leibniz-Institut DSMZ – Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH, Braunschweig

Prof. Dr. Yvonne Mast

Applied MicrobiologyAG Angewandte Mikrobiologie
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Research Topics

model organism: **actinobacteria** **model strain:** ***Streptomyces pristinaespiralis***

- Genome mining for natural products
- Investigation of regulatory mechanisms in antibiotic producing actinomycetes
- Activation of silent antibiotic gene clusters for the identification of novel natural compounds
- Genetic engineering of secondary metabolite biosynthetic pathways for generation of novel compound derivatives
- Synthetic Biology for novel compound production
- Optimization of production processes

Research Techniques

- genetic manipulation of actinomycetes (mutagenesis, heterologous expression, CRISPR-Cas)
- transcriptional analysis, report gene analysis
- protein expression/purification, electromobility shift assays (EMSA), *in vitro* transcription/translation analysis (ivTT), Bacterial-two-hybrid assays (BTH)
- antibacterial bioassays
- HPLC-MS analysis
- mutasynthesis
- bioinformatics: (antiSMASH, BigSCAPE, ARTS, TYGS)

Collaboration wishes

- access to clinically relevant and multidrug-resistant pathogens for antimicrobial activity testing and deposition in culture collection
- *in vitro* screening systems to evaluate activity of novel natural compounds and crude extracts
- collaboration in synthetic and natural product chemistry, compound isolation, modification, and structure elucidation
- exploration and validation of therapeutic applications for natural compounds (antibiotics and derivatives) bacteriophages, and phage-derived endolysins

Cell Physiology - Technische Universität Braunschweig

Prof. Dr. Jochen Meier

Institute for Cell and Neurobiology
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Research Topics

model organism: mouse **model tissue:** hippocampus

- structure-function relationships of the neurotransmitter receptor for glycine (GlyR)
- identification of novel ligands (agonists and antagonists) of the GlyR
- novel molecular tools for analyses of (non-canonical) RNA-splicing and Apobec1-dependent C-to-U RNA-editing at single cell resolution
- cell-autonomous counteract against pathological changes of RNA-splicing and RNA-editing
- microfluidic chambers for analysis of subcellularly compartmentalized neuronal function using multimodal characterization (electrophysiological, MEA, microscopy, respiration) of ectopic action potentials in primary neuron cultures and their relation to clinically relevant cases – Does the soma know what axons and presynapses are doing?

Research Techniques

- patch-clamp electrophysiology *in vitro* and *in vivo* (zebrafish)
- confocal microscopy, optogenetics
- primary neuron cell culture, neurons on (microfluidic) chip (with multielectrode array MEA)

Collaboration wishes

- application of our molecular tools for diagnostic and therapeutic purposes
- application of our multimodal microfluidic chambers / MEA to clinically relevant cases / model systems
- projects related to confocal microscopy and whole-cell patch-clamp electrophysiology for identification and/or verification of clinically relevant neurotransmitter receptor ligands
- computational molecular docking studies
- single cell RNA sequencing

Polymerchemie - TU Braunschweig

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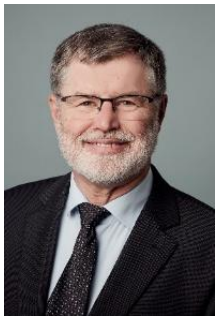

Research Topics

model organism: None **model tissue:** Bone, Gingiva,

- Drug delivery System,
  Nanogels, ultrathin coatings, enzyme triggered release
- Biofunctionalization of (dental implant) surfaces
  Ultrathin self-binding polymers, for drug delivery, antibacterial action while being compatible with human cells

Research Techniques

- Polymer synthesis and characterization
- Surface analytics (XPS, surface potential, …)
- Coating techniques (Spin Coating, dip coating, layer-by-layer deposition,…)
- Electrospinning

Collaboration wishes

- collaboration in the field of implant associated infections

Dept. of Systems Immunology - Helmholtz Centre for Infection Research
Prof. Dr. Michael Meyer-Hermann
Dr. Sebastian Binder

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Research Topics
Model organism: Virtual (in silico) human, animal and in vitro systems
Model tissue: Whole-body physiology

- Mathematical modelling and dynamical simulation of physiological systems in health and disease for hypothesis generation, testing and prediction.
- Adaptive control for therapy optimization: closed-loop strategies for personalized therapy and dynamic adjustment.
- Clinical decision support systems: mechanistic models + AI/ML for diagnosis, stratification, risk assessment and treatment decisions (e.g., post-/long-COVID, ME/CFS, chronic inflammation, ICU early warning systems).
- Immunology and vaccination: T-cell and germinal center dynamics, affinity maturation, antibody responses; data- and model-driven vaccine optimization.
- Pandemic dynamics and early warning: multi-scale models from within-host to population level; scenario analysis and public-health risk assessment.

Research Techniques

- Multiscale mathematical models and simulation of complex systems dynamics
- Parameter inference, model calibration/validation and optimization
- Adaptive control and machine learning for heterogeneous and noisy clinical data and therapy optimization

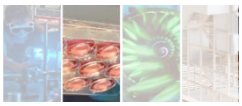
Collaboration wishes

- **Clinical partnerships & data:** Collaborations with clinical partners using structured longitudinal clinical data (incl. treatments and outcomes) for chronic conditions with unmet need and limited/ineffective standard therapies.
- **Complex, heterogeneous disease courses:** Applications with nonlinear dynamics and high inter-individual variability (e.g., fluctuating trajectories, responder/non-responder patterns), where modeling and simulation can support stratification, prediction, and treatment recommendations.

- **Decision support & closed-loop concepts:** Development of mechanistic/hybrid models (incl. digital twin–style approaches) for clinical decision support and adaptive/closed-loop treatment strategies (e.g., model-predictive adjustment).

- **Biomarkers under uncertainty:** Joint work on diagnostic, prognostic, and therapy-response biomarkers in settings with unclear mechanisms and difficult-to-interpret phenotypes.

Institut für Akustik und Dynamik- Technische Universität Braunschweig



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pvz Center of Pharmaceutical Engineering

Research Topics

model organism: diverse **model tissue:** diverse

- Modellbildung dynamischer Prozesse in Zellen (Alpha-Zellen, Beta-Zellen, Neuronen, Muskelzellen,…)
    - Elektrophysiologie und damit verknüpfte oszillatorische Phänomene
    - Ca2+ Ionen-Transport, Diffusion
    - Metabolismus
    - Ortsaufgelöste Modelle, Multiskalenmodelle
    - Phänomenologische Modellbildung, Fusions-/Fissionsprozesse von Organellen (z.B. Mitochondrien), Granulatransport
- Bilddatenverarbeitung von Mikroskopbildern
- Kontaktmechanik und Reibung, Tribologie, Verschleiß
- Flüsse durch enge Spalte
- Transport von Partikeln durch enge Spalte
- Schwingungen von Mehrkörpersystemen oder Strukturen
- Maschinendynamik, dynamische Materialcharakterisierung

Research Techniques

- **Modellbildung:**
    - Gewöhnliche Differentialgleichungen (Reaktion / Diffusion)
    - Partielle Differentialgleichungen, Finite Elemente, Finite Differenzen (Reaktion / Diffusion)
    - Zelluläre Automataen
- **Experimente:**
    - Diverse Bewegungs-sensierende Messsysteme
    - Mikroskope (konfokal, Weißlichtinterferometer)
    - diverse in-house Reibprüfstände

Collaboration wishes

- offen für jede Art von Fragestellungen / Kooperationen, in denen mathematische Modelle unterstützen bei
    - dem Verständnis zu komplexen dynamischen Wechselwirkungen in Zellen
    - der Entwicklung und Überprüfung von Hypothesen
    - Reproduktion / Präsentation von dynamischen Prozessen in Zellen
- Fragestellungen mit Bezug zu Fertigungsprozessen hinsichtlich der Maschinendynamik, Schwingungen und Materialcharakterisierung
- Fragestellungen zu Transportprozessen von Fluiden und Partikel durch enge Spalte
- Fragestellungen zu Kontaktmechanik und Reibung

TU Braunschweig, Psychotherapy and Diagnostics

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Research Topics

1. Work-related mental health problems (Work-Anxiety, Work Ability)
2. Neuropsychological dysfunctions (komplexe Teilleistungsstörungen, MCD, ADHS+ im Erwachsenenalter)
3. Coping with life problems
4. Behavior-oriented coaching and psychotherapy interventions, social medicine

Research Techniques

1. Psychopathology, clinical diagnostics
2. Psychopathology phenomenology
3. Employee surveys, clinical observation studies together with hospitals and practitioners
4. Longitudinal observation studies, randomized controlled intervention studies

Collaboration wishes

1. participants (patients, employees) can be sent to us for a **3-session coaching for allday and work problems:**

https://www.tu-braunschweig.de/psychologie/klinische/coaching

2. Physicians and psychotherapists working in outpatient care can participate in our **social medicine survey** (incentives: 200 €, social medicine glossary):

https://www.tu-braunschweig.de/psychologie/klinische/sozialmedizin-in-der-praxis

3. Open for **cooperation ideas on projects**, e.g.

- Behavior-oriented intervention research
- Side effects of psychotherapy
- Clinical phenomena: work-anxiety, work ability, neuropsychological dysfunctions (komplexe Teilleistungsstörungen, MCD, ADHS+ im Erwachsenenalter)

Standardization of Omics - Physikalisch-Technische Bundesanstalt/TUBS

Prof. Dr. Thomas Naake

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Research Topics

model organism: none **model tissue:** none

Research Techniques

- computational mass spectrometry (metabolomics, proteomics)
- uni- and multivariate statistical analysis, machine learning and deep learning, chemoinformatics
- standardization of metabolomics and proteomics
- quality assessment and control for (mass spectrometry) datasets

Collaboration wishes

- clinical or model organism omics datasets for single omics and multi-omics analysis, multi-omics integration and association to clinical phenotypes for improved understanding of molecular processes of disease subtypes, resistance mechanisms and vulnerabilities
- development of novel tools for omics data handling, analysis, and interpretation

Food Toxicology - Technische Universität Braunschweig

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Technische Universität Braunschweig

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Research Topics

model organism: mice, human **model tissue:** intestine, kidney

We study chemical reactions of proteins during food processing. We are particularly interested in the Maillard reaction, also called glycation, which describes the non-enzymatic reaction of sugars and amino compounds and contributes to flavor and aroma. Processed foods have been implicated in the development of many diseases, but the causal relationship of food processing and adverse health effects is very debatable. Data on the intake and effects of specific food components is often lacking.

Research Topics

- Characterization and quantification of protein modifications in model systems and food
- Impact of food processing and protein modification on protein digestibility
- Reactivity of food components during gastrointestinal digestion
- Effects of (digested) modified proteins on (gastrointestinal) epithelial cells
- Fermentation of modified proteins in food and in vivo
- Bioactivity of fermentation products

Research Techniques

-Analyses of protein and amino acid modifications via electrophoresis, HPLC-UV, LC-MS/MS
-Analysis of digestibility: static (Infogest), dynamic (bioreactor)
- Cell culture of human intestinal and renal cell lines
- Fermentation of modified proteins (yeasts, LAB, fecal samples)
- Enzyme activity assay
- Genotoxicity (Ames)

Collaboration wishes

- Biosamples → Analysis
- Interaction of microbiota and modified proteins
- Models for intestinal absorption
- Biomarkers for dietary intake

Institut für Pharmazeutische Technologie und Biopharmazie – Technische Universität Braunschweig

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Research Topics

model organism: **model tissue:**

- Primäre human-zellbasierte Systeme (z. B. Kornea-Epithelzellen, Nasenschleimhaut-Zellen, Hirnkapillar-Endothel)
- Immortalisierte Zelllinien (z. B. HCE-T, RPMI 2650, hCMEC/D3)
- 3D-Epithelbarrieren: cornealer und nasaler Gewebemodelle
- 3D-Endothelbarrieren: Blut-Hirn-Schranke (BBB) in vitro
- Mikrofluidische Organ-on-Chip-Systeme zur dynamischen Perfusion
- Tissue-Engineered Ocular Surface Constructs für regenerative Therapie
- Aufbau, Charakterisierung und Validierung von 3D-Barrieremodellen als Tiermodell-Alternative
- In vitro-Studien zu Arzneistoff-Absorption und Permeabilität
- Evaluierung neuartiger Drug-Delivery-Systeme und Tight-Junction-Modulatoren
- In vitro-Assessment der Bioäquivalenz pharmakologischer Formulierungen
- Integration von Mikrofluidik zur Verbesserung in vivo-ähnlicher Testbedingungen
- Tissue Engineering von Augenoberflächen für die okuläre Regenerationsmedizin

Research Techniques

- 3D-Zellkultur und Tissue Engineering (Scaffold- und Hydrogel-basierte Systeme)
- Zellcharakterisierung: konfokale Mikroskopie, Immunfluoreszenz, qPCR, Western Blot
- Drug-Permeationsassays: Transwell, Franz-Diffusionszellen, OOC-Chips mit TEER-Messung
- Analytische Verfahren: HPLC/LC-MS, radiolabeling, Fluoreszenz-Tracer
- Barrier Integrity Tests: Transendotheliale Elektrodenwiderstands-Messung, Marker-Permeabilität

Collaboration wishes

- Partnerschaften mit Pharmaunternehmen zur Validierung von 3D-Barrieremodellen
- Gemeinsame Entwicklung von Organ-on-Chip-Plattformen für Drug-Screening
- Kooperative Projekte zu Tight-Junction-Modulatoren und Bioverfügbarkeitsstudien
- Zusammenarbeit mit Klinik und Augenheilkundlern für regenerative Augenoberflächen

Austausch mit Regulierungsbehörden zur Etablierung in vitro-Bioäquivalenzrichtlinien

Institut für Pharmakologie, Toxikologie u. Klinische Pharmazie – Technische Universität Braunschweig

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Pharmaceutical Engineering

Research Topics

model organism:

- NMRI mouse
- SUR1 ko mouse
- NZO mouse

model tissue:

- islets of Langerhans (pancreatic islet)
- primary pancreatic beta cell
- primary pancreatic alpha cell
- insulin- or glucagon-secreting cell lines

question:

how is the metabolism of nutrients related to the regulation of hormone secretion?

Research Techniques

- isolation and culture of the above tissues
- transfection or viral transduction for the expression of fluorescent proteins
- perifusion of islets or pseudo-islets for the kinetic measurement of secretion
- conventional epifluorescence and TIRF microscopy
- rudimentary electrophysiology

Collaboration wishes

- CLEM would be interesting
- someone out there who is not only interested in but also experienced in dealing with voltage-dependent Ca2+ channels (Cav1.3)?

Biochemistry - Technische Universität Braunschweig

Prof. Dr. Anett Schallmey
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Research Topics

model organism: none **model tissue:** none

Enzymes as the broader research theme in my group, incl.
-Enzyme discovery using database mining, heterologous enzyme production, purification and enzyme characterization (stability, activity, kinetics, etc.)
-Enzyme engineering to improve characteristics such as enzyme stability, activity or selectivity, amongst others
-Enzyme immobilization
-Application of enzymes as biocatalysts, combination of enzymes in cascade reactions

Research Techniques

- Site-directed and site-saturation mutagenesis, high-throughput screening of enzyme mutant libraries incl. assay development
- Molecular dynamics simulations of proteins/enzymes
- Carrier-based covalent enzyme immobilization, preparation of cross-linked enzyme crystals (CLECs)
- Application of enzymes in batch and continuous-flow reactions, incl. microfluidics

Collaboration wishes

- I am open for any kind of collaboration in the context of enzymes.

Institut für Elektrische Messtechnik und Grundlagen der Elektrotechnik
TU Braunschweig

Prof. Dr. Meinhard Schilling

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Research Topics

Diagnostic methods:

- ECG and EEG
- Immunoassays
- MPI for imaging and theranostics
- MRI

Research Techniques

- multichannel ECG and EEG, as well with capacitive electrodes through clothes
- Preparation and functionalization of magnetic nanoparticles as markers – faster and cheaper than PCR and same sensitivity and specificity
- MRI and MPI imaging – functional imaging, as well for long term investigations
- Electron Microscopy – from bacteria to molecules

Collaboration wishes

- Functional and physiological imaging (CT, MRI, US,…)
- Immunoassays for identification of infections
- Geriatrics and sports medicine: Mobilisation after fall and surgeries
- We work together with Dr. Andreas Höft, Klinik für Anästhesiologie SKBS and during the last 25 years with Prof. Dr. Peter Werning

Molecular Infection Biology - Technische Universität Braunschweig

Prof. Dr. Michael Steinert
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Technische Universität Braunschweig

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Research Topics

model organism: **human** **model tissue:** **lung tissue explants**

- Host-pathogen interaction and pathomechanisms of Legionnaires´disease Macrophages, interactome, Peptidyl-prolyl *cis/trans* isomerases, Zinc metalloproteases
- Virulence blockers and tissue protectives FK506 analogues

Research Techniques

-Human lung tissue explants

-Cell culture of macrophages, protozoa, nematodes

-Cellular infections (S2)
-Molecular tool box
-Interactomics
-Cytotoxicity, antiinfectives

Collaboration wishes

-tissue scanning

-AI in histology

Health-Care Architecture and health protection - Technische Universität Braunschweig

Prof. Dr. Wolfgang Sunder
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Research Topics

-Topic 1: Interdisciplinary planning of healthcare buildings
-Topic 2: Built environment and infection, prevention and control
-Topic 3: Built environment and the health and well-being
-Topic 4: Resilient hospital architecture
-Topic 5: Structural and technical heat protection in hospitals
-Topic 6: Hospital architecture in low- and middle-income countries
-Topic 7: Healthy city

Research Techniques

-Technique 1: Design and construction of prototype areas within the hospital (Rooms, fittings, façade, building)
-Technique 2: Digital Planning Tools & Simulation
-Technique 3: Participatory planning, involvement of hospital user groups
-Technique 4: Qualitative and quantitative survey
-Technique 5: Monitoring of areas within the hospital (e.g. via sensors)
-Technique 6: Literature Review

Collaboration wishes

Collaboration wishes

-collaboration proposal 1: Survey of user groups (doctors, nurses, logistics staff, cleaning staff) in various areas of the hospital

-collaboration proposal 2: Assessment of existing buildings and new-build projects for resilience (IPC, natural disasters, crises)

-collaboration proposal 3: Development of training modules for clinicians on Build Environment and infection, prevention and control

Cellular and Molecular Neurobiology - Technische Universität Braunschweig

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Research Topics

model organism: zebrafish, mouse **model tissue:** cerebellum, brain, limb

- molecular and cellular mechanisms of mammalian and zebrafish development
- investigation of genetic models of human diseases
- limb development (mouse)
- spinocerebellar ataxias (SCAs 1, 2, 3) (mouse, zebrafish)
- behavioral analysis (mouse, zebrafish)
- embryonic development of genetic models

Research Techniques and Expertise

- genetic manipulation of mouse embryonic stem cells (ES)
- in vitro differentiation of ES-cells
- knockout and gene trap mutagenic approaches
- gene expression analysis
- whole-mount in situ hybridization
- genetic manipulation of zebrafish
- cell culture and cell signaling

Collaboration wishes

- modelling of human neurological diseases in zebrafish and mouse
- testing therapeutic approaches for neurological diseases in animal models
- neurological diseases and behavioral read-out (behavioral analysis)

Biophysical Chemistry - Technische Universität Braunschweig
Prof. Dr. Peter Jomo Walla

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Research Topics

model (organism): **liposome models, mice, zebrafish, plants (depending on cooperation partners)** **model tissue:** **single proteins up to entire tissues**

- biophysical mechanisms of membrane interactions: docking, hemifusion and full membrane fusion processes, neuronal signaling, immune cell membrane interactions, virus entry, exocytosis

- molecular mechanisms of neuronal signal transmission or virus–host interactions at membranes: receptor binding, membrane deformation, fusion intermediates and their kinetics, nanoscale membrane distance measurements

- membrane-mediated protein interactions and cytoskeleton coupling: role of actin dynamics in membrane remodeling, synaptic spine growth, protein complex formation at membranes

- single-molecule dynamics of proteins in cellular and viral systems: nanoscale tracking of receptors, viral proteins and membrane components during interaction and signaling processes

Research Techniques

- advanced fluorescence spectroscopy: fluorescence correlation (FCS) and cross-correlation spectroscopy (FCCS), fluorescence lifetime microscopy

- single molecule and super-resolution imaging: single particle tracking, high-resolution fluorescence microscopy (<50 nm), single protein dynamics

- optical tweezers combined with fluorescence microscopy: manipulation of cells, viruses and liposomes to study membrane interactions and interaction forces

- liposome model systems for membrane biophysics: reconstituted GUV and SUV protein systems to study docking, full fusion, nanoscale membrane distances

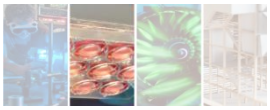
- two-photon microscopy and live-cell fluorescence imaging

Collaboration wishes

- Endocytosis of mRNA-vaccines and -drugs into cells and tissue, potentially human – how does the entry at the membranes work exactly on a molecular level?

- functionalized chambers for multimodal characterization of primary neuronal cultures, comparison to clinical samples (similar and in cooperation as project proposed by R. Köster)

Institut für Pharmazeutische Biologie - Technische Universität Braunschweig



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pvz Center of Pharmaceutical Engineering

Research Topics

model organism: plants /plant cells **model tissue:**

- bioactive natural products (specialized metabolites) from plants, especially glucosinolates
- structural diversity, biological roles and evolution of specialized metabolites
- specialized metabolites in organismic interactions (plants-insects, plants-microbes)
- biotechnological production of bioactive natural products in heterologous systems
- metabolic engineering & synthetic biology
- plant suspension cultures as production platforms for bioactive compounds
- new bioactives fromplant-associated organisms

Research Techniques

- genetic transformation of plants and plant suspension cultures for reconstruction of biosynthesis pathways using multigene constructs
- genome editing in plants
- small molecule analytics (HPLC, GC-MS, HPLC-MS)
- heterologous expression of enzymes in Arabidopsis, tobacco, *E. coli*, *S. cerevisiae*, *P. pastoris*, or Sf9 cells
- enzyme purification and characterization

Collaboration wishes

- therapeutic proteins to be produced in plant-based production platforms
- gut model to test effects plant specialized metabolites on microbiome
- bioactivity tests